

Integrating Agentic Workflows

Scaling Productivity for Uber's Driver Onboarding Team

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Agenda

- 1. Introduction & Domain Landscape**
- 2. Developer Workspace Skills**
- 3. Tooling & Platform Architecture**
- 4. Major Case Study & Demo: Driver Test Account AI Agent**
- 5. Impact & Key Takeaways**

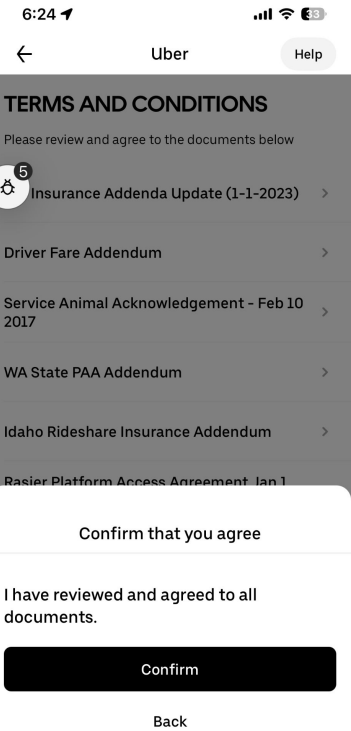
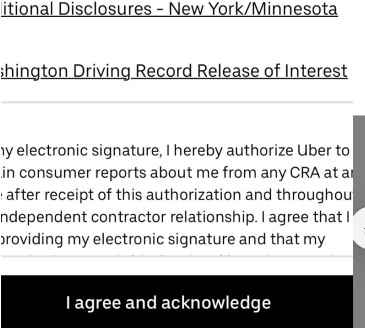
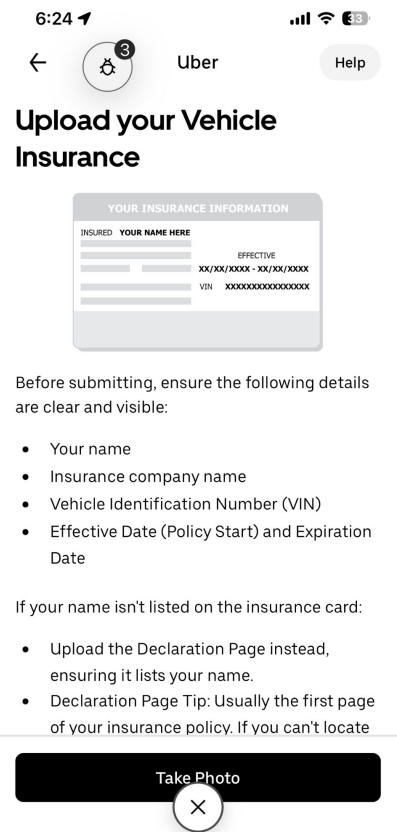
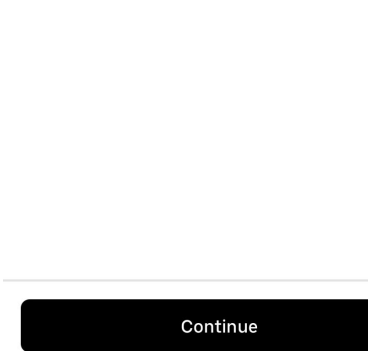
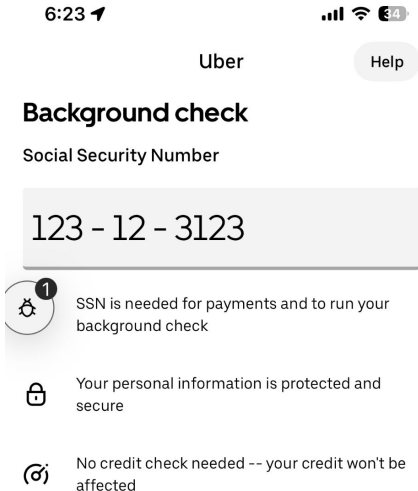
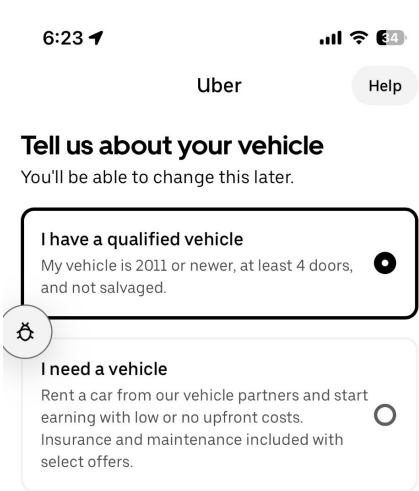
Driver Onboarding

Compliance & Screenings

Earner Lifecycle Management

Multi-Channel Onboarding

Identity & Document Verification



Friction: Enabling True Agentic Development with Monorepo Scale



2,200+ Services

Massive unified monorepo causes checkout lag, indexing slowdowns, and frequent local config drift during branch switching.

Fix: Isolated /worktree Automation

Targeted Monorepo Sparse-Checkout

The custom `/worktree` command isolates the Driver Onboarding sub-tree instantly, eliminating indexing overhead and branch collisions.

- ✓ Zero full-repo re-indexing penalties
- ✓ Parallel multi-task development across clean workspaces

```
# Automated workspace isolation
$ /worktree ONBOARD-4921
[git] Creating worktree at .worktrees/ONBOARD-4921
[git] Applying sparse-checkout: src/driver-onboarding/*

✓ Isolated tree ready in 1.4s (saved 98% disk/index)
```

Friction: Strict CI Gates



Strict CI Gatekeeping

Rigid PR templates and mandatory test evidence blocks cause verbose, generic LLM diffs to fail formatting gates repeatedly.

Fix: Branch-to-Diff Skill

Branch-to-Diff Rule

Extracts Jira context, verifies local unit test coverage, and formats PR summaries matching team guidelines to pass CI on attempt #1.

Pass CI validation on first attempt

Friction: Knowledge Drift



Knowledge Drift

Outdated documentation leads code models to generate obsolete testing signatures and deprecated mock configurations.

Fix: EngWiki Sync Skill

EngWiki Doc Sync

Synchronizes Confluence and internal guides directly with updated service schemas, eliminating outdated patterns in LLM completions.

Ground truth documentation for agents

Friction: Administrative Toil



Manual JIRA Tracking

Engineers spend valuable sprint time manually creating subtasks, filling required namespace fields, and synchronizing ticket status after diff merges

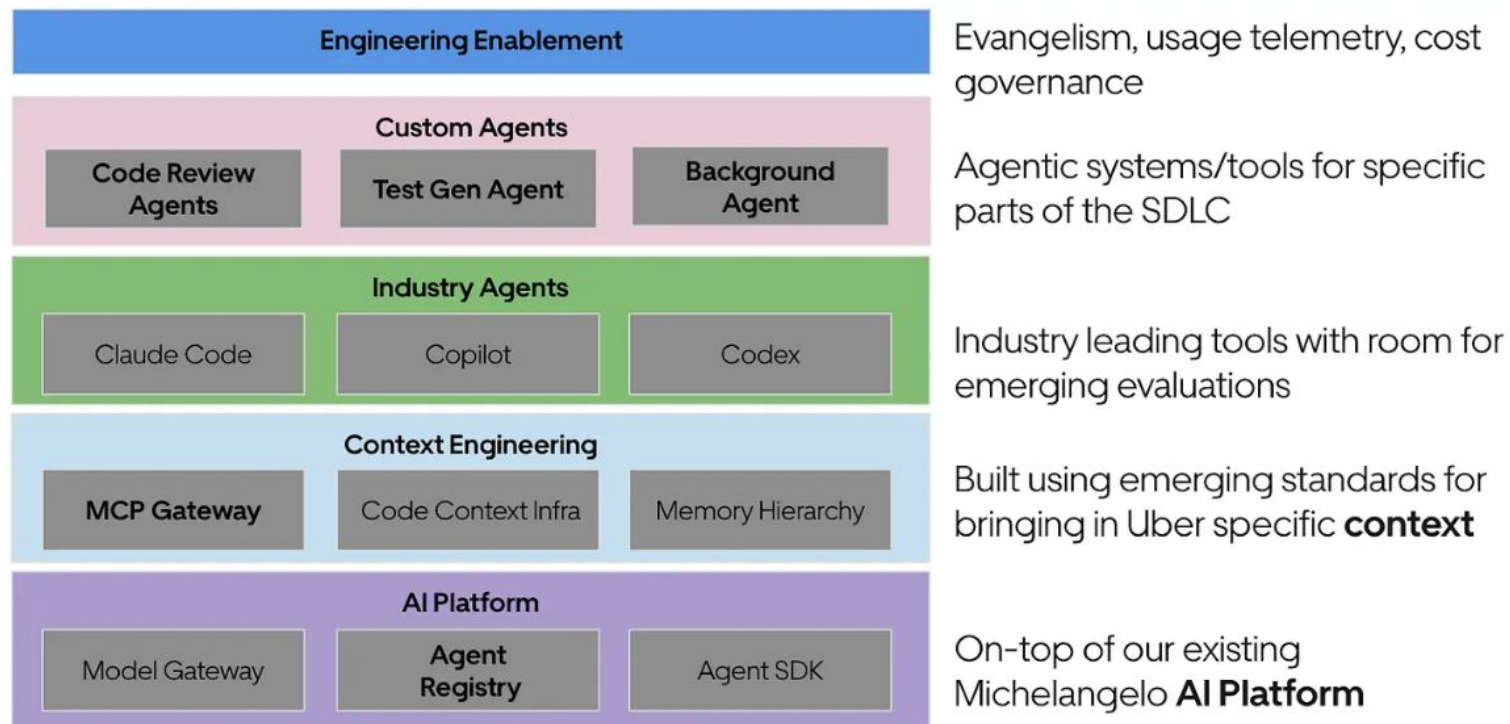
Fix: JIRA REST Skill

JIRA REST Skill

Automated ticket creation with team namespaces, pulls acceptance criteria into prompts, and transitions ticket states

Uber's AI Platform & Ecosystem

Agentic Layer Cake



Source: Pragmatic Engineer

**Agent
Builder**

**MCP
Gateway**

Agent Builder

- No Code Agent
- Native Tooling & MCP Integration
- Dynamic Sub-Agent Handoffs

Instructions * ⓘ

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You are an alert assistant that has access to:

- [sawmill.anamoly_detection.error_logs_query](#) for checking logs of a service
- [up.up_sdk.get_service_operations](#) for getting history of operations on a service. Only show the Deploy and Restart ones.
- [flipr.property.get_flipr_history](#) for getting changes in history

You can also handoff to the following expert agents:

- [elaina-test-on-call-log-analysis__override](#) for log analysis
- @|

tool

Type

agent

Own

{x} variable

micelangelo-generative-ai



Context knowledge

Supported file types: pdf json txt

Upload files

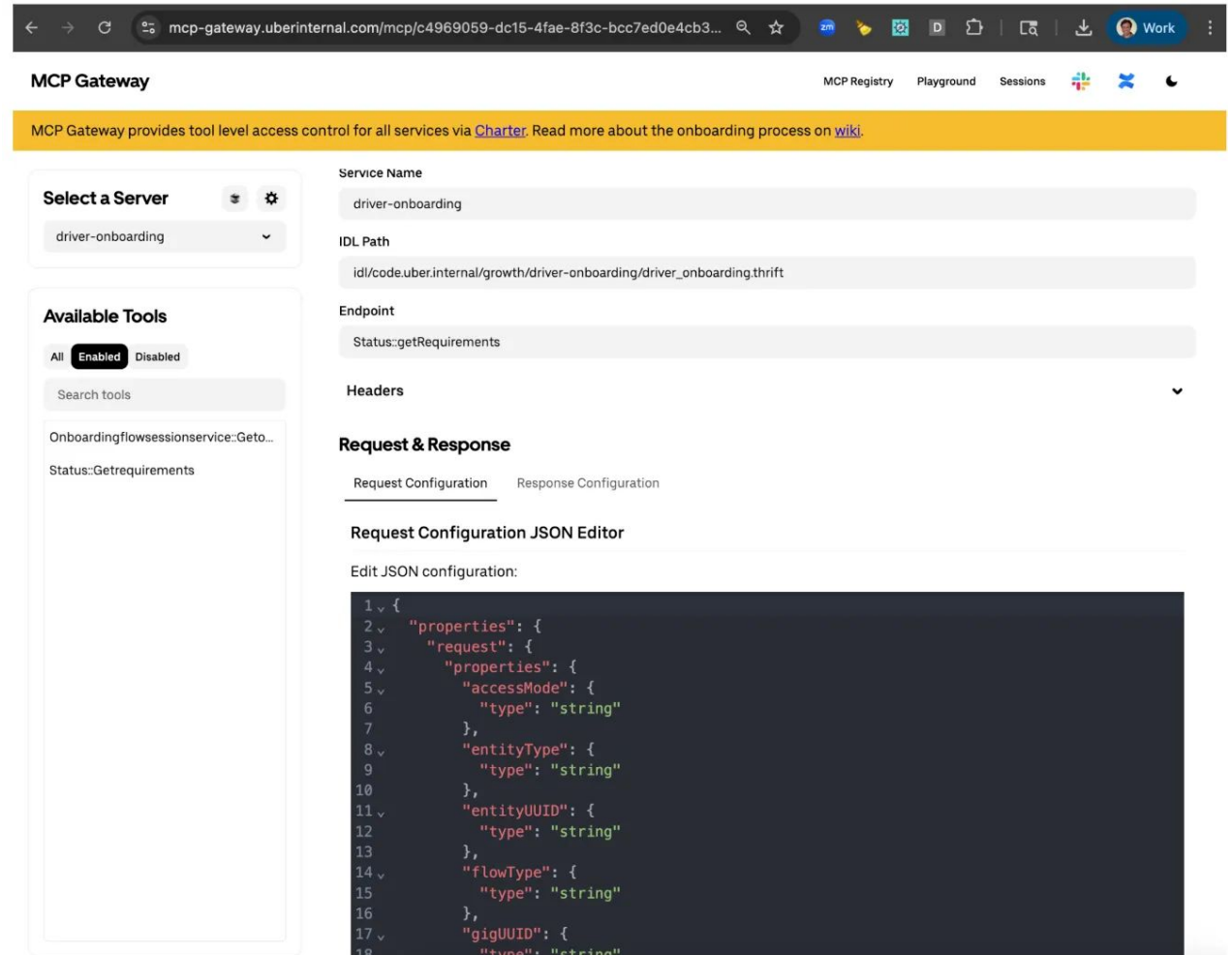
Browse files

or drop to upload

Source: Pragmatic Engineer

MCP Gateway

- Exposes Core Microservices as Tools
- Decouples Agent Logic from Service APIs
- Built in Observability & Audit Trails



Test Account AI Agent

15,000+ cities

San Francisco, Seattle, Accra,
London, NYC, Bangalore...

15+ Background Check Error States

Insufficient Licensing History,
failed SSN, General Rejection...

20+ earning opportunities

Car Seat, Alcohol Delivery,
Eats, Rides, Tasks, Commercial
Rides...

10+ Document Rejection States

Poor Picture Quality, Wrong
Document, Document Blurry...

Vehicle Insurance States

Terms and Conditions states

Various Surfaces

New features..



50,000+ Unique Onboarding States

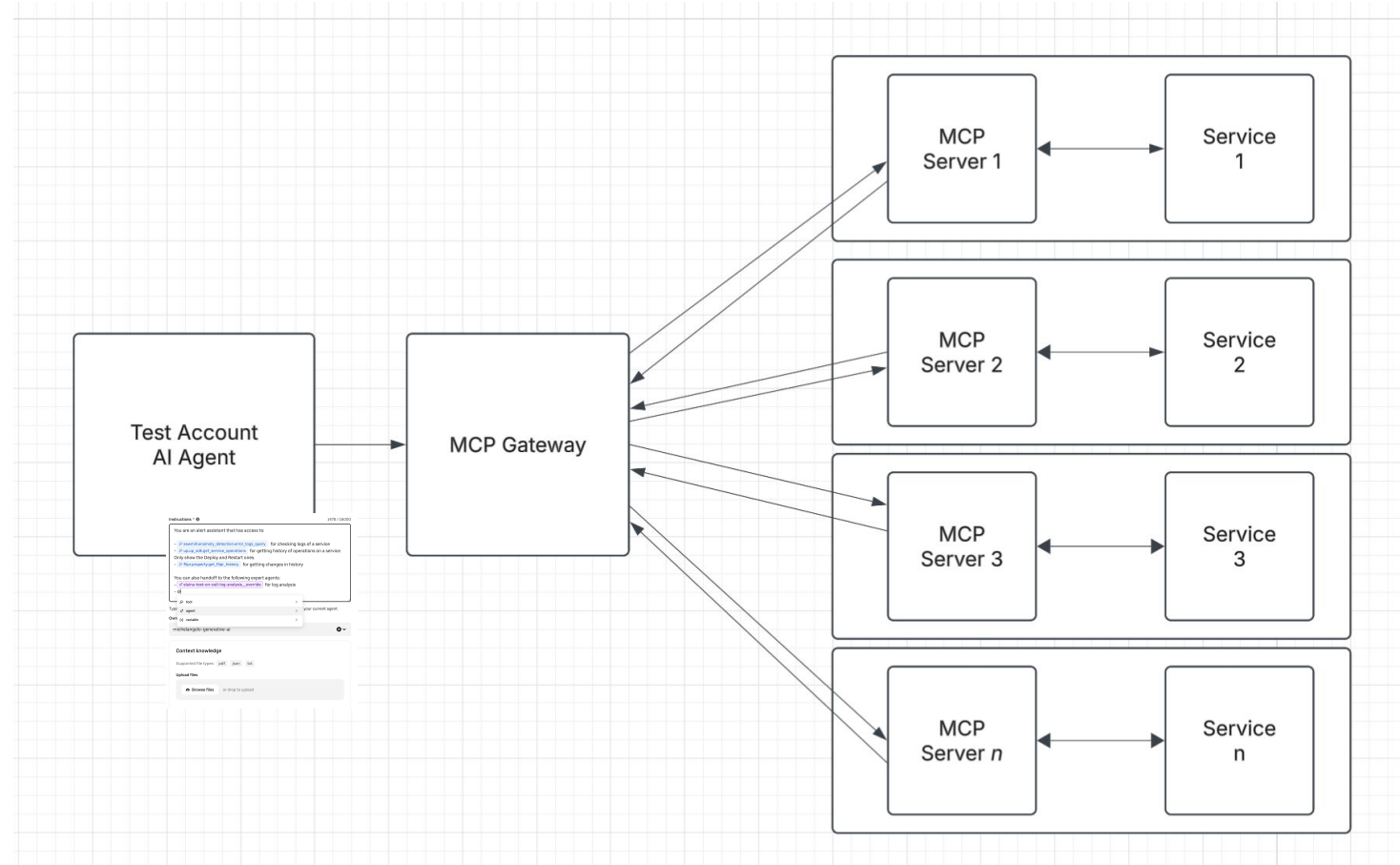
Uploaded Driver License, Rejected
Profile Photo, Expired Document,
Incomplete T&Cs

Test Account AI Agent - Internals

Create a delivery test account in Seattle with an approved driver's license but failed background check

Phase 1: Create Test Account

Phase 2: Transition Test Account (to proper state)



Quantifiable Business & Engineering Impact

<30s

Setup Duration

Down from 5–10 minutes

~95%

Creation Success Rate

Across benchmark test suite

100%

Self-Serve Access

Via Slack BMO Bot for QA & Ops

Takeaways

01

Decouple Reasoning from State Mutation

Use agents for dynamic discovery, but hand off execution to atomic batch engines.

02

Let Agents Work in Parallel

Use git worktrees or other mechanisms so multiple agents can work at the same time without blocking each other.

03

Benchmark Tool Calls, Not Vibe Checks

Skip subjective LLM judges; test agent reliability using deterministic tool selection accuracy and task coverage.

04

Safety Belongs in Code, Not Prompts

Never rely on prompt instructions for security; block invalid mutations with deterministic backend interceptors.

THANK YOU!

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